OPERATIONS MANAGEMENT

USING OPERATIONS TO COMPETE

- **Operations management** refers to the systematic design, direction, and control of processes that transform inputs into services and products for internal, as well as external customers. OM can be a source of competitive advantage for firms in both service as well as manufacturing sectors. This particularly deals with managing those fundamental activities and processes that organizations use to produce goods and services that people use every day.

- **A process** is any activity or group of activities that takes one or more inputs, transforms them, and provides one or more outputs for its customers. For organizational purposes, processes tend to be clustered together into operations.

- **An operation** is a group of resources performing all or part of one or more processes.

- Processes can be linked together to form a **supply chain**, which is the interrelated series of processes within a firm and across different firms that produce a service or product to the satisfaction of the customers.

- **Supply chain management** is the synchronization of a firm's processes with those of its suppliers and customers to match the flow of materials, services, and information with customer demand. For example, a book printing company must schedule the printing of a very large quantity of books in a timely fashion, receive orders from its largest customers, directly load and dispatch a fleet of trucks by specific destination while bypassing regular warehouses, keep track of their progress using technology, and finally, bill their customers and collect payment. Sound operational planning, along with internal and external coordination within its supply chain, provides one example of designing customized processes for competitive operations and creates wealth & value for a firm's stake holders.

Role of Operations in an Organization

- Operations and supply chain management underlie all departments and functions in a business.

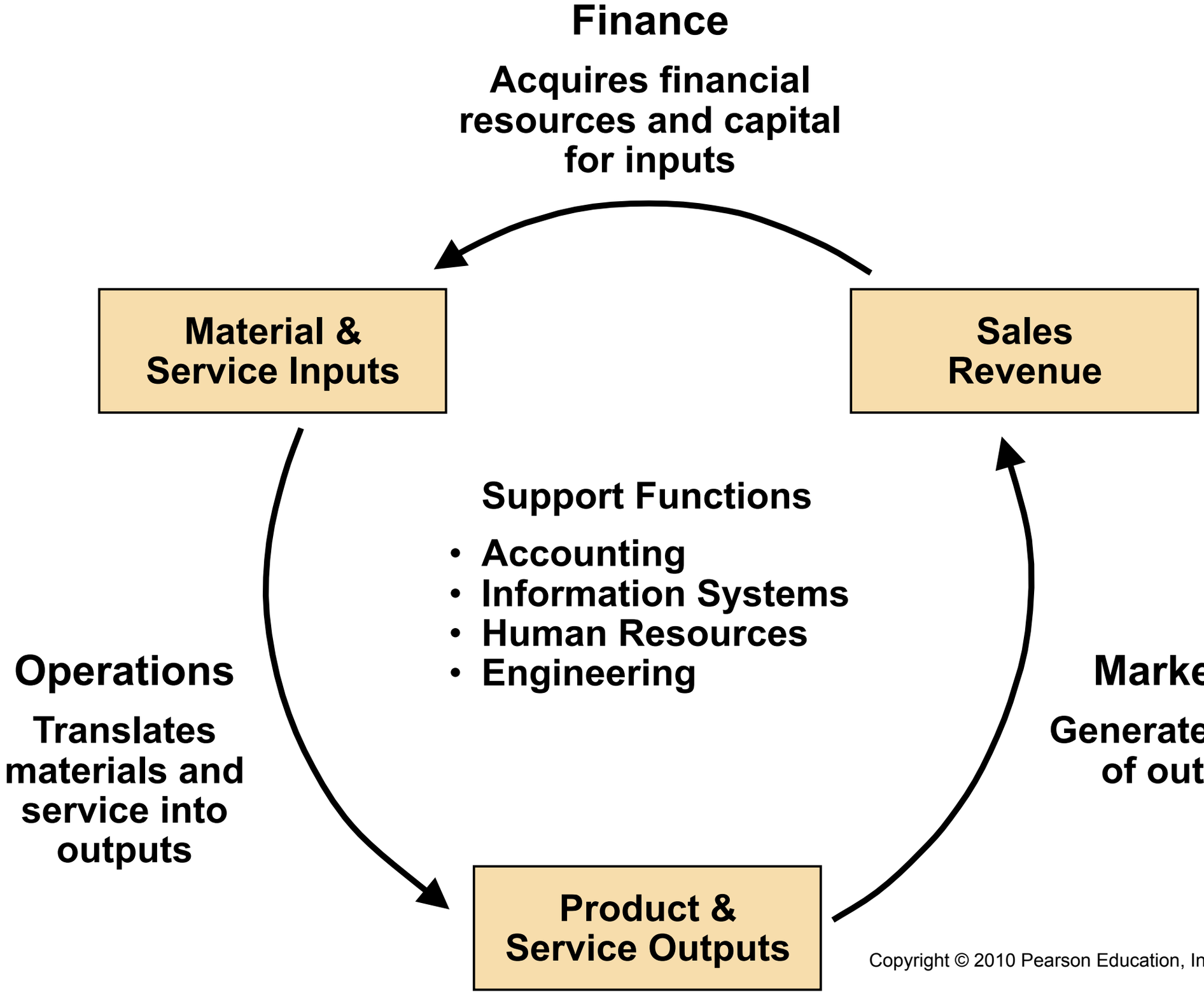
- Operations serve as an excellent career path to upper management positions in many organizations.

- The reason is that operations managers are responsible for key decisions that affect the success of the organization. (Ex: Chief Operations Officer - COO). Reporting to the head of operations are the managers of departments, such as customer service, production and inventory control, and quality assurance.

- The three mainline functions of any business are 1. Operations, 2. Marketing, and 3. Finance. Each function is unique and has its own knowledge and skill areas, primary responsibilities, processes, and decision domains. Functions such as accounting, information systems, human resources, and engineering make the firm complete by providing essential information, services, and other managerial support.

- These functions & relationships provide direction for the business as a whole, and are integrated & aligned to the same strategic intent. How well these functions work together determines the effectiveness of the organization. Regardless of how departments and functions are individually managed; they are always linked together through processes.

- Thus, a firm competes not only by offering new services and products, creative marketing, and skillful finance, but also through its unique competencies in operations and sound management of core processes.



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A Process View

- You might wonder why we begin by looking at processes, rather than at departments or even the firm. The reason is that a process view of the firm provides a much more relevant picture of the way firms actually work.

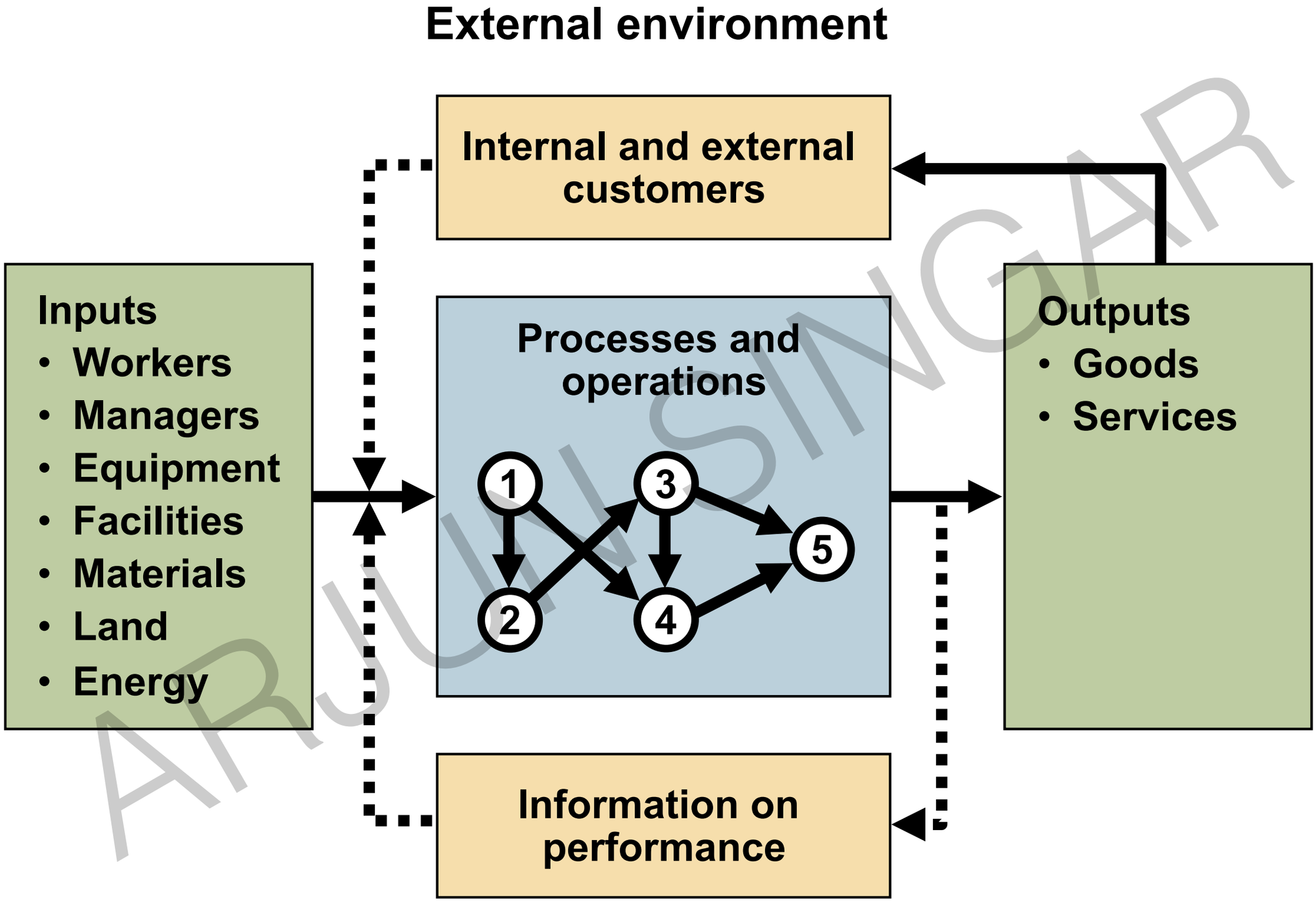
- Departments typically have their own set of objectives, a set of resources with capabilities to achieve those objectives, and managers and employees responsible for performance.

- A process can have its own set of objectives, involve a work flow that cuts across departmental boundaries, and require resources from several departments.

- The key to success in many organizations is a keen understanding of how their processes work, since an organization is only as effective as its processes.

- Every process and every person in an organization has customers and also relies on suppliers.

- Therefore, operations management is relevant and important for all students, regardless of major, because all departments have processes that must be managed effectively to gain a competitive advantage.



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How Processes Work

- Any process has inputs and outputs. Inputs can include a combination of human resources (workers and managers), capital (equipment and facilities), purchased materials and services, land, and energy.

- Processes provide outputs to customers. These outputs may often be services (that can take the form of information) or tangible products. Every process and every person in an organization has customers and suppliers.

- **External Customers** are customers who are either end users or intermediaries (ex: manufacturers or retailer) buying the firm's finished services or products.

- **Internal Customers** are employees or processes that rely on inputs from other employees or processes to perform their work.

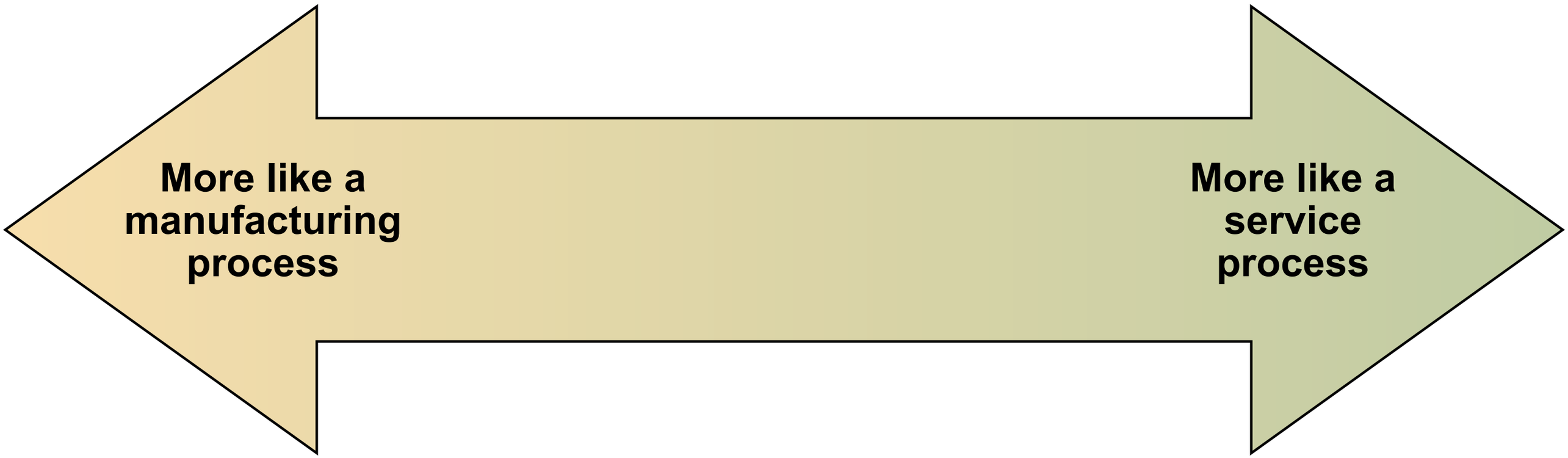
- **External Suppliers** are other businesses or individuals who provide the resources, services, products, and materials for the firm's short term and long term needs.

- **Internal Suppliers** are the employees or processes that supply important information or materials to a firm's processes.

- **Note:** Processes can be broken down into subprocesses, which in turn can be broken down further into still more subprocesses. We refer to this concept of a process within a process as a **nested process**.

Service and Manufacturing Processes

- The two major types of processes are 1. Service and 2. Manufacturing.
- Service processes pervade the business world that take the form of information. Manufacturing processes, without them, the products we enjoy as part of our daily lives would not exist. In addition, manufacturing gives rise to service opportunities.
- Their **two key differences** are 1. the nature of their output and 2. the degree of customer contact.
- **Manufacturing** processes convert materials into goods that have a physical form which we call products. In general, manufacturing processes also have longer response times, are more capital intensive, and their quality can be measured more easily than those of service processes. The outputs from manufacturing processes can be produced, stored, and transported.
- If a process does not change the physical properties or shape or size or assembly of the material, it is considered a **service**. Service processes tend to produce intangible, perishable outputs.
- Service processes tend to have a higher degree of customer contact than the manufacturing processes. Customers may take an active role in the process itself whereas manufacturing processes tend to have less customer contact. the important point is that managers must recognize the degree of customer contact required when designing processes.
- With **regards to similarities**, at the level of the firm, service providers do not just offer services (ex: restaurant) and manufacturers do not just offer products (ex: buying a computer).
- Further, **service processes do inventory their inputs** but not finished goods inventories (ex: hospitals) and **some manufacturing processes do not inventory their outputs** because they are too costly (ex: tailored suits).
- However, there are few organizations that are both manufacturers as well as service providers (Ex: Burger Joints).



- Physical, durable output
- Output can be inventoried
- Low customer contact
- Long response time
- Capital intensive
- Quality easily measured

- Intangible, perishable output
- Output cannot be inventoried
- High customer contact
- Short response time
- Labor intensive
- Quality not easily measured

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A Supply Chain View

- Most services or products are produced through a series of interrelated business activities.

- Each activity in a process should add value to the preceding activities; waste and unnecessary cost should be eliminated.

- Our process view of a firm is helpful for understanding how services or products are produced and why cross-functional coordination is important, but it does not shed any light on the strategic benefits of the processes. The missing strategic insight is that processes must add value for customers throughout the supply chain.

- Here, the concept of supply chains reinforces the link between processes and performance, which includes a firm's internal processes as well as those of its external customers and suppliers.

- There are essentially, two main types of processes in the supply chain, namely, 1. Core Processes and 2. Support Processes.

- **Core Processes:** A core process is a set of activities that delivers value to external customers.

- Managers of these processes and their employees interact with external customers and build relationships with them, develop new services and products, interact with external suppliers, and produce the service or product for the external customer.

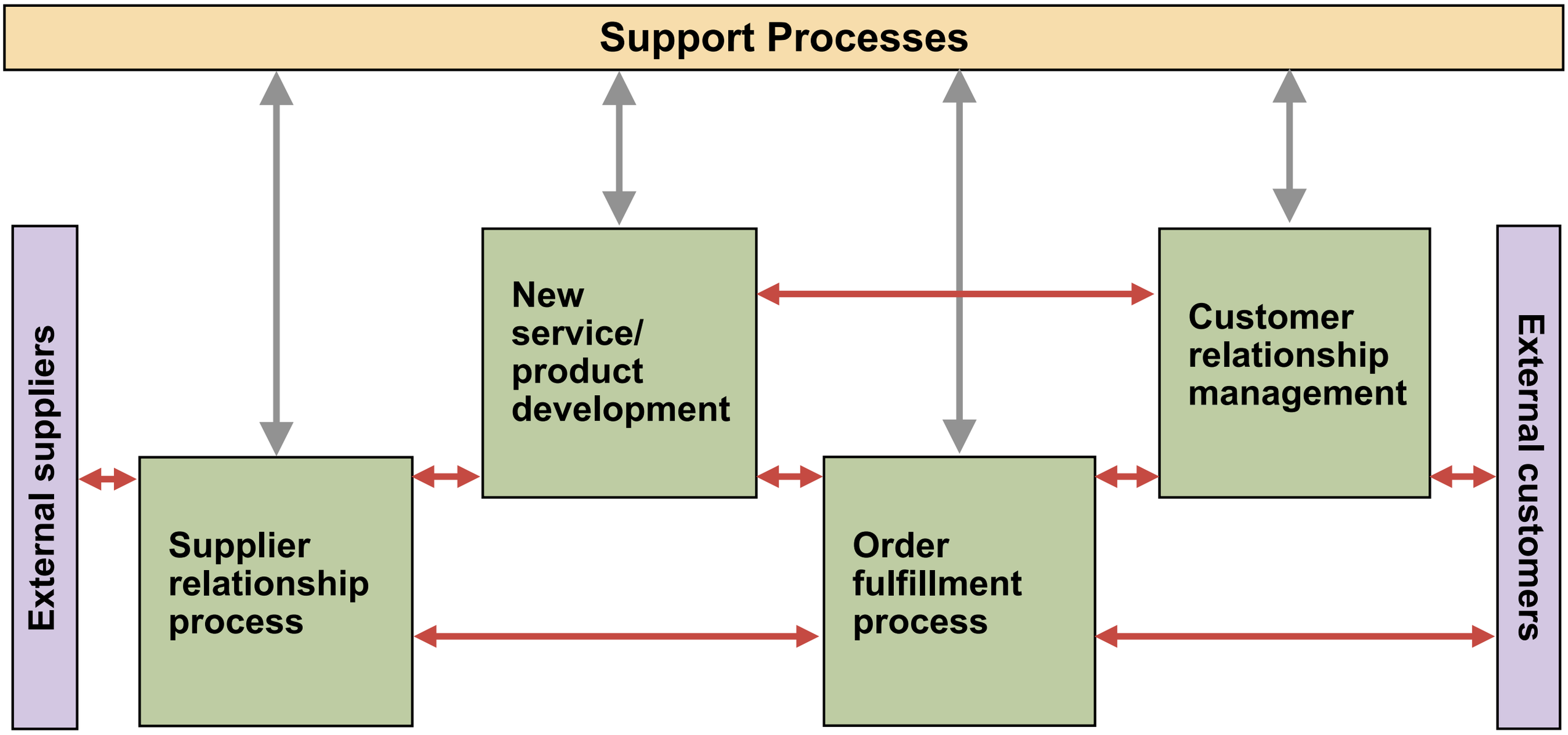
- Example include a hotel's reservation handling, A new car design for an auto manufacturer, or Web-based purchasing for an online retailer like Amazon).

1. Supplier Relationship Process: A process that selects the suppliers of services, materials, and information and facilitates the timely and efficient flow of these items into the firm. (Ex: Negotiating fair prices, Scheduling on-time deliveries).

2. New Service / Product Development Process: A process that designs and develops new services or products from inputs received from external customer specifications or from the market in general through the customer relationship process. (Ex: Prototypes - Smart phones).

3. Order Fulfillment Process: A process that includes the activities required to produce and deliver the service or product to the external customer. (Ex: Packaging and Transportation or Electronic - Mode of Delivery).

4. Customer Relationship Process / CRM: A process that identifies, attracts, and builds relationships with external customers and facilitates the placement of orders by customers, sometimes referred to as customer relationship management. (Ex: Marketing and Sales, Customer Database).



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- **Support Processes:** A support process provides vital resources and inputs to the core processes and is essential to the management of the business.

- Firms have many support processes. Processes as such are not just in operations but are found in accounting, finance, human resources, management information systems, and marketing.

- Examples include budgeting, recruiting, and scheduling. Support processes provide key resources, capabilities, or other inputs that allow the core processes to function.

- The Human Resources function in an organization provides many support processes such as recruiting and hiring workers who are needed at different levels of the organization, training the workers for skills and knowledge needed to properly execute their assigned responsibilities, and establishing incentive and compensation plans that reward employees for their performance.

- The legal department puts in place support processes that ensure that the firm is in compliance with the rules and regulations under which the business operates.

- The Accounting function supports processes that track how the firm's financial resources are being created and allocated over time, while the Information Systems function is responsible for the movement and processing of data and information needed to make business decisions.

- All of these support processes must be managed to create as much value for the firm and its customers.

- Managers of these processes must understand that they cut across the organization, regardless of whether the firm is organized along functional, product, regional, or process lines (Cross-functional coordination).

TABLE 1.1 | EXAMPLES OF SUPPORT PROCESSES	
Capital acquisition	The provision of financial resources for the organization to do its work and to execute its strategy
Budgeting	The process of deciding how funds will be allocated over a period of time
Recruitment and hiring	The acquisition of people to do the work of the organization
Evaluation and compensation	The assessment and payment of people for the work and value they provide to the company
Human resource support and development	The preparation of people for their current jobs and future skills and knowledge needs
Regulatory compliance	The processes that ensure that the company is meeting all laws and legal obligations
Information systems	The movement and processing of data and information to expedite business operations and decisions
Enterprise and functional management	The systems and activities that provide strategic direction and ensure effective execution of the work of the business

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Supply Chain Business Processes

- **Outsourcing:** Exploring available suppliers for the best options to perform processes in terms of price, quality, delivery, time, environmental issues.

- **Warehousing:** Receiving shipments from suppliers, verifying quality, placing in inventory, and reporting receipt for inventory records.

- **Sourcing:** Selecting, certifying, and evaluating suppliers and managing supplier contracts.

- **Customer Service:** Providing information to answer questions resolve problems using automated information services as well as voice-to-voice contact with customers.

- **Logistics:** Selecting transportation mode, scheduling both inbound and outbound shipments, and providing intermediate inventory storage.

- **Cross-docking:** Packing of products of incoming shipments so they can be easily sorted more economically at intermediate warehouses for outgoing shipments to their final destination.

Operations Strategy

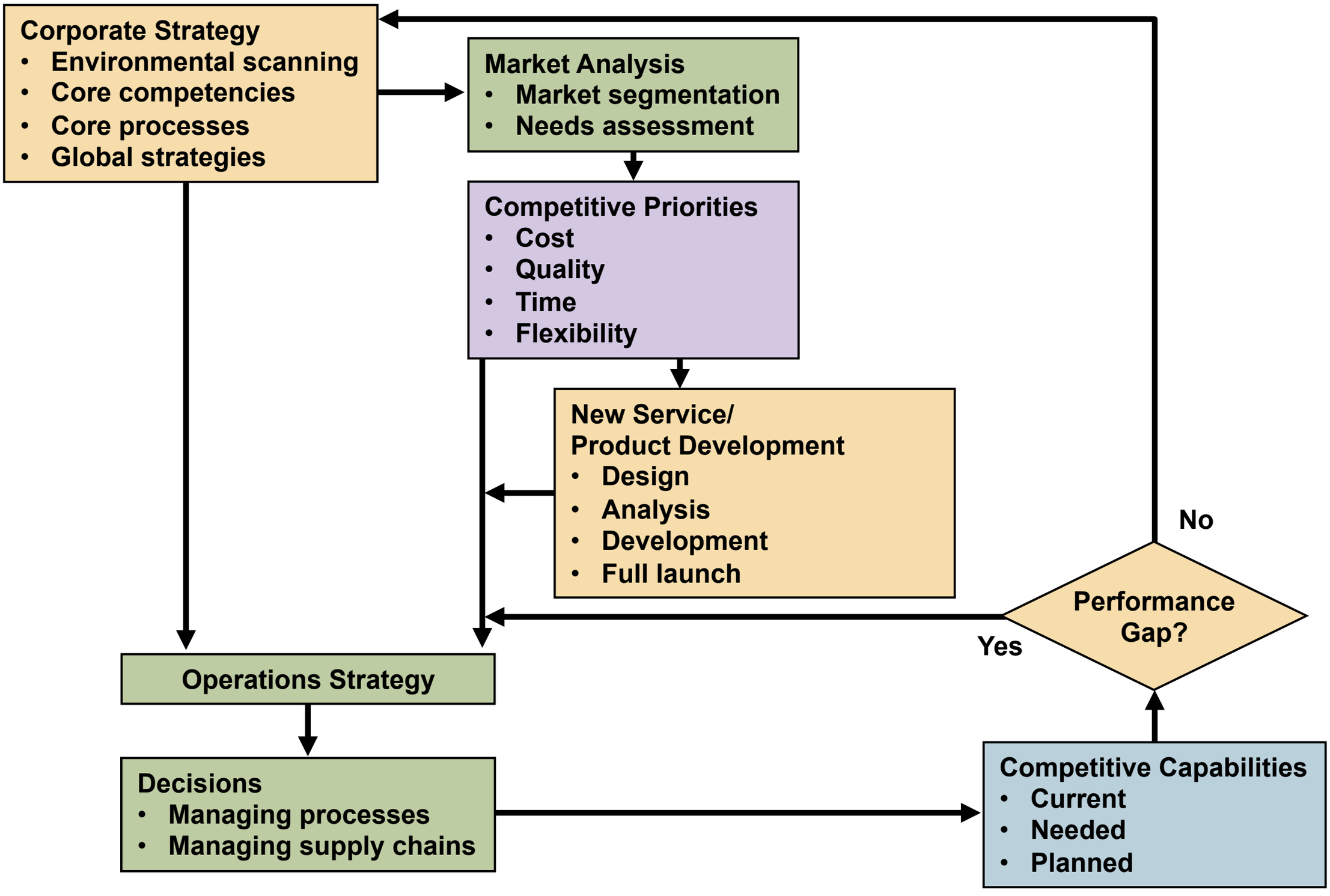
- Operations strategy specifies the means by which operations implements corporate strategy and helps to build a customer-driven firm.

- OS is typically used to develop the capabilities the firm needs to be competitive. This is the heart of managing processes and supply chains.

- A firm’s internal processes which are the building blocks, need to be organized to ultimately be effective in a competitive environment.

- Hence, this is the linchpin that brings these processes together to form supply chains encompassing suppliers as well as customers.

- Since customers constantly desire change, the firm’s operations strategy must be driven by the needs of its customers.



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Corporate Strategy

- Developing a customer-driven operations strategy begins with corporate strategy which coordinates the firm's overall goals with its core processes.

- Corporate strategy provides an overall direction that serves as the framework for carrying out all the organization's functions. It specifies the businesses the company will pursue, isolates new opportunities and threats in the environment, and identifies growth objectives. Developing CS involves four considerations:

1. Environmental Scanning: Monitoring and adjusting to the changes in the business environment; for opportunities and threats. (Ex: Electric powered cars to replace gasoline).

2. Developing Core Competencies: They are unique resources and strengths that an organization's management considers when formulating strategy. They basically reflect on how to coordinate processes and integrate technologies. These competencies include - **i.** A well trained and flexible workforce, **ii.** Well-located facilities, **iii.** Market and Financial Know-How, and **iv.** Implementation of Systems and Technology.

3. Developing Core Processes: A firm's core competencies should drive its core processes - customer relationship, new service or product development, order fulfillment, and supplier relationship. (Ex: Banking, Health-Care).

4. Developing Global Strategies: This may include buying foreign service and parts, combating threats from foreign competitors, or planning ways to enter markets beyond traditional national boundaries. Two effective global strategies are **i.** strategic alliance (collaboration, joint venture and technology licensing) and **ii.** location abroad. (Ex: Hotels, Education, Automobiles, QSRs).

Market Analysis

- Market Analysis helps to understand what the customer wants and how to provide it. It first divide's the firm's customers into market segments and then identifies the needs of each segment.

1. Market Segmentation: This is the process of identifying groups of customers with enough in common to warrant the design and provision of services or products that the group wants and needs. To identify market segments, the analyst must determine the characteristics that clearly differentiate each segment. (Ex: Gap M&W vs Gap Kids and Future Group's Fashion @ Big Bazaar for lower middle class vs Pantaloon for upper middle class).

2. Needs Assessment: This identifies the needs of each segment and assesses how well competitors are addressing those needs. Market needs should include both the tangible and intangible attributes and features of products and services that a customer desires. These needs are; **i.** Service or Product Needs - Price, quality and degree of customization, **ii.** Delivery System Needs - Convenience, courtesy, safety, accuracy, reliability, delivery speed and dependability, **iii.** Volume Needs - High or low volume and degree of variability and predictability in volume, and **iv.** Other Needs - After sale technical support, ability to invest in global markets, and competent legal services.

Competitive Priorities and Capabilities

A customer-driven operations strategy requires a cross-functional effort by all areas of the firm to understand the needs of the firm's external customers and to specify the operating capabilities the firm requires to outperform its competitors. Such a strategy also addresses the needs of internal customers because the overall performance of the firm depends upon the performance of its core and supporting processes, which must be coordinated to provide the overall desirable outcome for the external customer.

- **Competitive priorities** are the critical operational dimensions a process or supply chain must possess to satisfy internal or external customers, both now and in the future. They must be present to maintain or build market share or to allow other internal processes to be successful. Not all competitive priorities are critical for a given process; management selects those that are most important.

- **Competitive capabilities** are the cost, quality, time, and flexibility dimensions that a process or supply chain actually possesses and is able to deliver. When the capability falls short of the priority attached to it, management must find ways to close the gap or else revise the priority.

- Here we focus on nine broad competitive priorities that fall into the four capability groups of cost, quality, time, and flexibility.

TABLE 1.2 | DEFINITIONS, PROCESS CONSIDERATIONS, AND EXAMPLES OF COMPETITIVE PRIORITIES			
COST	**Definition**	**Process Considerations**	**Example**
1. Low-cost operations	Delivering a service or a product at the lowest possible cost	Processes must be designed and operated to make them efficient	Costco
QUALITY			
2. Top quality	Delivering an outstanding service or product	May require a high level of customer contact and may require superior product features	Ferrari
3. Consistent quality	Producing services or products that meet design specifications on a consistent basis	Processes designed and monitored to reduce errors and prevent defects	McDonald's
TIME			
4. Delivery speed	Quickly filling a customer's order	Design processes to reduce lead time	Dell
5. On-time delivery	Meeting delivery-time promises	Planning processes to increase percent of customer orders shipped when promised	United Parcel Service (UPS)
6. Development speed	Quickly introducing a new science or a product	Cross-functional integration and involvement of critical external suppliers	Li & Fung

TABLE 1.2 | DEFINITIONS, PROCESS CONSIDERATIONS, AND EXAMPLES OF COMPETITIVE PRIORITIES

FLEXIBILITY	Definition	Process Considerations	Example
7. Customization	Satisfying the unique needs of each customer by changing service or products designs	Low volume, close customer contact, and easily reconfigured	Ritz Carlton
8. Variety	Handling a wide assortment of services or products efficiently	Capable of larger volumes than processes supporting customization	Amazon.com
9. Volume flexibility	Accelerating or decelerating the rate of production of service or products quickly to handle large fluctuations in demand	Processes must be designed for excess capacity	The United States Postal Service (USPS)

- Management assigns selected competitive priorities to each process that are consistent with the needs of external as well as internal customers. Competitive priorities may change over time.

- For example, consider a high-volume standardized product, such as color ink-jet desktop printers. In the early stages of the ramp-up period when the printers had just entered the mass market, the manufacturing processes required consistent quality, delivery speed, and volume flexibility. In the later stages of the ramp-up when demand was high, the competitive priorities became low-cost operations, consistent quality, and on-time delivery.

- Competitive priorities must change and evolve over time along with changing business conditions and customer preferences.

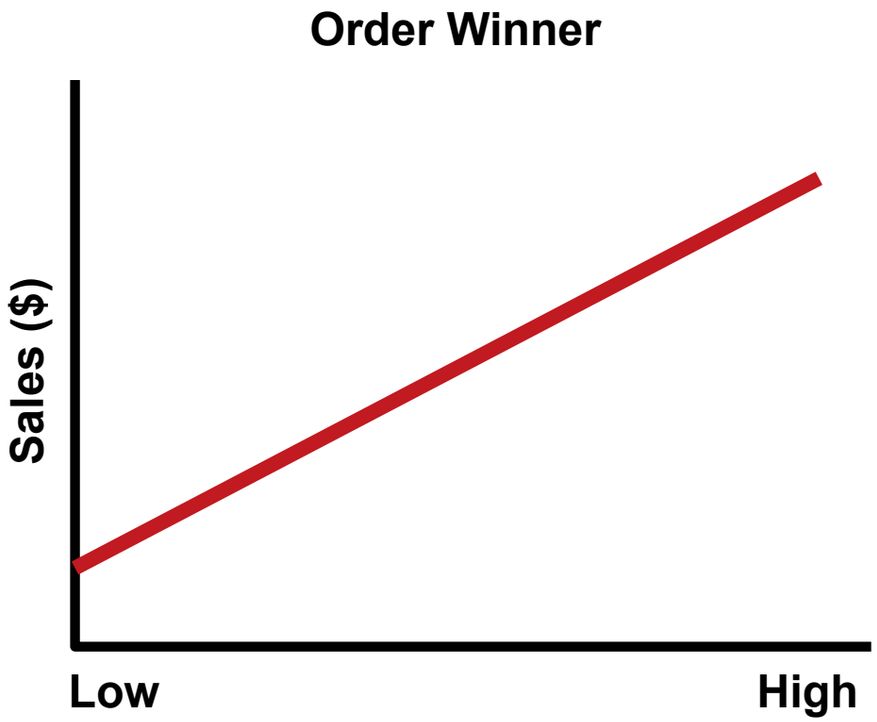
Order Winners and Order Qualifiers

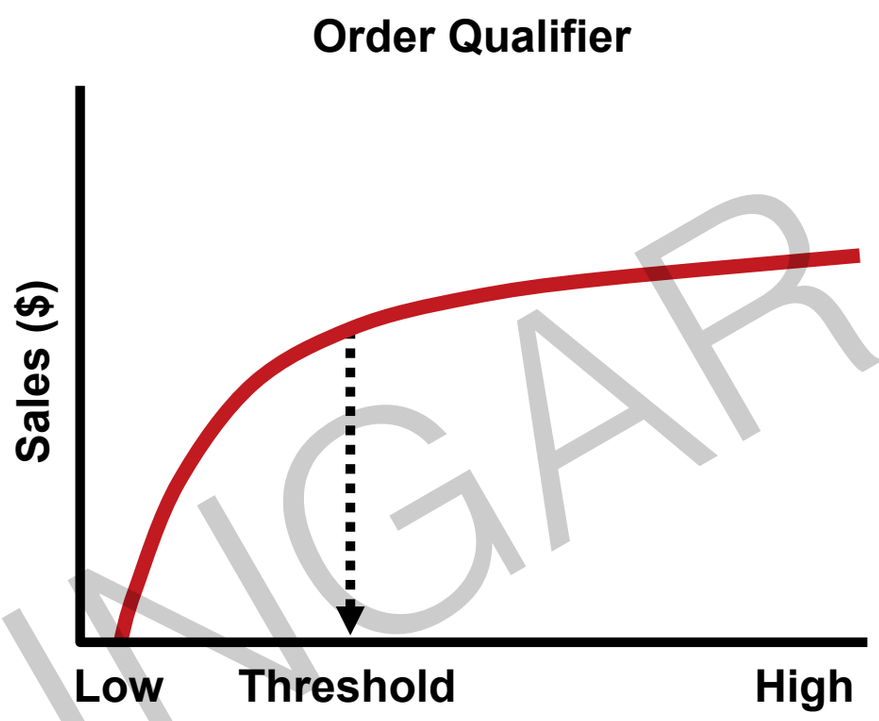
Competitive priorities focus on what operations can do to help a firm be more competitive, and are in response to what the market wants. Another useful way to examine a firm's ability to be successful in the marketplace is to identify the order winners and order qualifiers.

- An **order winner** is a criterion that customers use to differentiate the services or products of one firm from those of another. Order winners can include price (low-cost operations), quality, time, and flexibility. However, it may also include after-sale support, technical support, company reputation, and customer satisfaction. Order winners are derived from the considerations customers use when deciding which firm to purchase a service or product from in a given market segment. (ex: low-cost).

- Sometimes customers demand a certain level of demonstrated performance before even contemplating a service or product. Minimal level required from a set of criteria for a firm to do business in a particular market segment is called **order qualifier**. (ex: consistent quality).

- Fulfilling the order qualifier will not ensure competitive success; it will only position the firm to compete in the market. From an operations perspective, **understanding which competitive priorities are order qualifiers and which ones are order winners is important** for the investments made in the design and management of processes and supply chains.





- The above figures show how order winners and qualifiers are related to achieving the competitive priorities of a firm.

- For example, an **order-qualifying** dimension could be **consistent quality** (Air Asia) and **order-winning** dimension could be low price driven by **low-cost** operations (Toyota Corolla and Honda Civic).

- We will consider two market segments: (1) first-class passengers and (2) coach passengers. Core services for both market segments are ticketing and seat selection, baggage handling, and transportation to the customer's destination. Here the competitive priorities for the first-class segment are top quality and on-time delivery, whereas the competitive priorities for the coach segment are low-cost operations, consistent quality, and on-time delivery.

ARJUN SINGAR

Using Competitive Priorities: An Airline Example

TABLE 1.2 | COMPETITIVE PRIORITIES ACROSS DIFFERENT CORE PROCESSES FOR AN AIRLINE

	CORE PROCESSES			
Priority	**Supplier Relationship**	**New Service Development**	**Order Fulfillment**	**Customer Relationship**
Low Cost Operations	Costs of acquiring inputs must be kept to a minimum to allow for competitive pricing.		Airlines compete on price and must keep operating costs in check.	
Top Quality		New services must be carefully designed because the future of the airline industry depends on them.	High quality meal and beverage service delivered by experienced cabin attendants ensures that the service provided to first-class passengers is kept top notch.	High levels of customer contact and lounge service for the first-class passengers.
Consistent Quality	Quality of the inputs must adhere to the required specifications. In addition, information provided to suppliers must be accurate.		Once the quality level is set, it is important to achieve it every time.	The information and service must be error free.
Delivery Speed				Customers want immediate information regarding flight schedules and other ticketing information.
On time delivery	Inputs must be delivered to tight schedules.		The airline strives to arrive at destinations on schedule, otherwise passengers might miss connections to other flights.	
Development Speed		It is important to get to the market fast to preempt the competition.		
Customization		The process must be able to create unique services.		
Variety	Many different inputs must be acquired, including maintenance items, meals and beverages.		Maintenance operations are required for a variety of aircraft models.	The process must be capable of handling the service needs of all market segments and promotional programs.
Volume Flexibility	The process must be able to handle variations in supply quantities efficiently.			

Operations Management as a set of Decisions and Addressing the Challenges of OM

- In practice, managers make strategic and tactical decisions
- Each part of the organization designs and operates processes
- Each function is connected through shared resources



Trends and Challenges in Operations Management

- Productivity Improvement
- Global Competition
- Ethical, Workforce Diversity and Environmental Issues

Productivity Improvement

- **Productivity** is a basic measure of performance for economies, industries, firms, and processes. It is the value of outputs (services and products) produced divided by the values of input resources (wages, cost of equipment, and so on) used.

- **Measuring Productivity:** Many measures are available. For example, value of output can be measured by what the customer pays or simply by the number of units produced or customers served. The value of inputs can be judged by their cost or simply by the number of hours worked.

- Labor Productivity = Index of the output per person or per hour worked.

- Similar measures may be used for machine productivity, where the denominator is the number of machines.

- Multi-factor Productivity = Index of the output provided by more than one of the resources used in production; it may be the value of the output divided by the sum of labor, materials, and overhead costs.

EXAMPLE 1.1

Calculate the productivity for the following operations:

a. Three employees process 600 insurance policies in a week. They work 8 hours per day, 5 days per week.

SOLUTION

$$\text{a. Labor productivity} = \frac{\text{Policies processed}}{\text{Employee hours}}$$

$$= \frac{600 \text{ policies}}{(3 \text{ employees})(40 \text{ hours/employee})} = 5 \text{ policies/hour}$$

EXAMPLE 1.1

Calculate the productivity for the following operations:

b. A team of workers makes 400 units of a product, which is sold in the market for $10 each. The accounting department reports that for this job the actual costs are $400 for labor, $1,000 for materials, and $300 for overhead.

SOLUTION

a. Multifactor productivity = $\dfrac{\text{Value of output}}{\text{Labor cost + Materials cost + Overhead cost}}$

$$= \frac{(400 \text{ units})(\$10/\text{unit})}{\$400 + \$1{,}000 + \$300} = \frac{\$4{,}000}{\$1{,}700} = 2.35$$

Solved Problem 1

Student tuition at Boehring University is $150 per semester credit hour. The state supplements school revenue by $100 per semester credit hour. Average class size for a typical 3-credit course is 50 students. Labor costs are $4,000 per class, material costs are $20 per student per class, and overhead costs are $25,000 per class.

a. What is the *multi-factor* productivity ratio for this course process?

b. If instructors work an average of 14 hours per week for 16 weeks for each 3-credit class of 50 students, what is the *labor* productivity ratio?

Solved Problem 1

SOLUTION

a. Multifactor productivity is the ratio of the value of output to the value of input resources.

$$\text{Value of output} = \left(\frac{\text{50 student}}{\text{class}}\right)\left(\frac{\text{3 credit hours}}{\text{student}}\right)\left(\frac{\text{\$150 tuition + \$100 state support}}{\text{credit hour}}\right)$$

$$= \$37{,}500/\text{class}$$

$$\text{Value of inputs} = \text{Labor} + \text{Materials} + \text{Overhead}$$

$$= \$4{,}000 + (\$20/\text{student} \times 50 \text{ students/class}) + \$25{,}000$$

$$= \$30{,}000/\text{class}$$

$$\text{Multifactor productivity} = \frac{\text{Output}}{\text{Input}} = \frac{\$37{,}500/\text{class}}{\$30{,}000/\text{class}} = 1.25$$

Solved Problem 1

SOLUTION

b. Labor productivity is the ratio of the value of output to labor hours. The value of output is the same as in part (a), or $37,500, so

$$\text{Labor hours of input} = \left(\frac{14 \text{ hours}}{\text{week}}\right)\left(\frac{16 \text{ weeks}}{\text{class}}\right)$$

$$= 224 \text{ hours/class}$$

$$\text{Labor productivity} = \frac{\text{Output}}{\text{Input}} = \frac{\$37{,}500/\text{class}}{224 \text{ hours/class}}$$

$$= \$167.41/\text{hour}$$

Solved Problem 2

Natalie Attire makes fashionable garments. During a particular week employees worked 360 hours to produce a batch of 132 garments, of which 52 were "seconds" (meaning that they were flawed). Seconds are sold for $90 each at Attire's Factory Outlet Store. The remaining 80 garments are sold to retail distribution at $200 each. What is the *labor* productivity ratio of this manufacturing process?

Solved Problem 2

SOLUTION

$$\text{Value of output} = (52 \text{ defective} \times 90/\text{defective}) + (80 \text{ garments} \times 200/\text{garment})$$

$$= \$20,680$$

$$\text{Labor hours of input} = 360 \text{ hours}$$

$$\text{Labor productivity} = \frac{\text{Output}}{\text{Input}} = \frac{\$20,680}{360 \text{ hours}}$$

$$= \$57.44 \text{ in sales per hour}$$

Application

	This Year	Last Year	Year Before Last
Factory unit sales	2,762,103	2,475,738	2,175,447
Employment (hrs)	112,000	113,000	115,00
Sales of manufactured products ($)	$49,363	$40,831	—
Total manufacturing cost of sales ($)	$39,000	$33,000	—

- Calculate the year-to-date labor productivity:

	This Year	Last Year	Year Before Last
$\frac{\textit{factory unit sales}}{\textit{employment}}$	$\frac{2,762,103}{112,000} = 24.66/hr$	$\frac{2,475,738}{113,000} = 21.91/hr$	$\frac{2,175,447}{115,000} = \$18.91/hr$

- Calculate the multifactor productivity:

	This Year	Last Year
$\frac{\textit{sales of mfg products}}{\textit{total mfg cost}}$	$\frac{\$49,363}{\$39,000} = 1.27$	$\frac{\$40,831}{\$33,000} = 1.24$